

ITAI 1374 – Midterm Project

Telco Customer Churn

Group Contribution Journal

Group Members:

Patrick Ray

Viviana Zamora

Funsho Orogun

Dustin Mastrota

Individual Contribution – Patrick Ray

I downloaded the original Telco Customer Churn dataset from Kaggle and organized the project files. I uploaded the dataset and group notebook to Google Colab, where I independently executed and verified the preprocessing notebook from beginning to end.

I reviewed the 70/30 training and testing split, missing-value handling, feature engineering, categorical encoding, numerical scaling, class balancing, visualizations, and final integrity checks. I also downloaded and organized the completed notebook and cleaned training dataset for our group submission.

Through this project, I learned why exploratory data analysis should be performed only on the training data and why the testing dataset must remain untouched. I gained a better understanding of how missing values are handled, how categorical information is converted into numerical features, and why class balancing can be useful when preparing data for machine learning.

My contribution also includes coordinating the group, collecting team members' files, organizing the final documents, and preparing the materials for the GitHub submission.

Individual Contribution – Viviana Zamora

I selected Telco Customer Churn as our project topic and contributed to the Python code and data cleaning. The group is using portions of my Jupyter notebook as part of our shared project.

I also helped revise the supporting documents and team reflection, including correcting the instructor approval date.

Through this work, I learned why data cleaning should be done in code and why testing data must remain separate from training-data analysis.

Evidence of my work can be found in the data-cleaning and preprocessing sections of 03_eda_preprocessing.ipynb, the cleaned output in 06_cleaned_training_data.csv, and the revised reflection in 04_reflection_and_contributions.pdf.

Individual Contribution – Dustin Mastrota (Levi)

My contribution to our Telco Customer Churn midterm project was to complete the analysis and review portion. I reviewed the dataset source document, project proposal, and EDA/preprocessing notebook to understand our project's goal and how the data was prepared.

I then wrote a technical review explaining the major preprocessing steps, including handling missing values, encoding categorical features, scaling numerical features, creating additional features, and balancing the classes. My review focused on why these steps were important and how they prepare the dataset for future machine learning models.

Throughout the project, I stayed in contact with my teammates to understand what each person contributed and how their work fit into the overall project. This helped me write a review that accurately explained our team's work while keeping my own role clear.

My responsibility was to review and explain the preprocessing rather than develop the original notebook code or train a model.

Individual Contribution – Funsho Orogun

I removed unnecessary data such as customerID, handled missing values in TotalCharges, converted categorical data into numerical values, and scaled numerical features. I also used SMOTE to balance the churn classes, which increased the dataset size and made it more suitable for training. Overall, I learnt how data cleaning and preparation can improve the quality of a machine learning dataset.